Chapter 1. Time Series Data

Dependence, stationarity and autocovariance

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1.1 Time series data with dependence

Motivation. Observations recorded over time are correlated with their own past, so the i.i.d. toolkit must be replaced by models for dependence.

1.1.1 Trend and moving-average smoothing

For observations $x_1, \dots, x_n$ (e.g. monthly stock index values), the *moving average* of span m is

$$y_t = \frac{x_{t-m+1} + \cdots + x_t}{m}.$$

Larger m gives a smoother curve (*smoothing*): short spans show short-run movement, long spans the trend.

Under the signal-plus-noise model

$$x_t = f(t) + z_t, \qquad t = 1, \dots, n, \tag{1.1}$$

with f deterministic and z_t i.i.d. $(0, \sigma^2)$,

$$y_t = \frac{f(t-m+1) + \cdots + f(t)}{m} + \frac{z_{t-m+1} + \cdots + z_t}{m}. \tag{1.2}$$

The second term has variance σ^2/m and is $\approx Ez_1 = 0$ by the LLN, so $y_t \approx f_m(t) := m^{-1} \sum_{j=0}^{m-1} f(t-j)$.

1.1.2 Exponential smoothing

Discounting the past geometrically, for $0 < w < 1$,

$$S_t = w \sum_{j=0}^{t-1} (1-w)^j x_{t-j}.$$

Given S_0^*, define recursively

$$S_t^* = wx_t + (1-w)S_{t-1}^*. \tag{1.3}$$

Iterating,

$$S_t^* = w \sum_{j=0}^{t-1} (1-w)^j x_{t-j} + (1-w)^t S_0^* = S_t + (1-w)^t S_0^*, \tag{1.4}$$

so $S_t^* \approx S_t$ for large t. Take $S_0^* = x_1$ or $\bar{x}$, and choose w to minimize

$$SSE(w) = \sum_{t=2}^{n} (x_t - S_{t-1}^*)^2. \tag{1.5}$$

1.1.3 Trend, seasonality and noise

With trend $T_t = f(t)$, the decomposition $x_t = T_t + S_t + z_t$ has *seasonal component* $S_t = S_{t+s}$ ($s = 4$ quarterly, $s = 12$ monthly). Real errors z_t are usually correlated, which motivates stationarity.

1.2 Stationary time series

Motivation. Inference from one realization needs a probabilistic structure that is stable over time; its dependence is described by the autocovariance function.

Definition 1.2.1 (Time series)

If T is the set of integers $\mathbb{Z}$ or of positive integers $\mathbb{Z}^+ = \{1, 2, \dots\}$, a collection of random variables $\{X_t : t \in T\}$ is called a *time series*.

Definition 1.2.2 (Strictly stationary time series)

$\{X_t\}$ is *strictly stationary* if for all $k \geq 1$, all integers $t_1, \dots, t_k$ and h, and all real $x_1, \dots, x_k$,

$$P(X_{t_1} \leq x_1, \dots, X_{t_k} \leq x_k) = P(X_{t_1+h} \leq x_1, \dots, X_{t_k+h} \leq x_k).$$

Definition 1.2.3 (Weakly stationary time series)

$\{X_t\}$ is *(weakly) stationary* if
(1) $EX_t = \mu$ for all t;
(2) $EX_t^2 < \infty$ for all t;
(3) $\mathrm{Cov}(X_t, X_{t+h}) = \mathrm{Cov}(X_0, X_h)$ for all t, h.

By (3) the covariance depends only on the lag; in particular the variance is constant. "Stationary" means weakly stationary.

Example 1.2.4 (Strict versus weak stationarity)

(1) An i.i.d. sequence $\{X_t\}$ is strictly stationary.
(2) If moreover $EX_1^2 < \infty$, it is also weakly stationary. An i.i.d. Cauchy sequence is strictly but not weakly stationary.
(3) If ϵ_t are i.i.d. with mean 0, variance 1, then $X_t = \epsilon_t + \epsilon_{t-1}$ is strictly and weakly stationary.

A strictly stationary series with finite variance is weakly stationary (shift invariance of the law of (X_t, X_{t+h}) gives (1) and (3)); finite variance is assumed throughout.

Definition 1.2.5 (Autocovariance function)

For a stationary $\{X_t\}$ the real function

$$\gamma(h) = \mathrm{Cov}(X_t, X_{t+h}), \qquad h = 0, \pm 1, \pm 2, \dots,$$

is called the *autocovariance function* (ACVF).

Theorem 1.2.6 (Basic properties of the ACVF)

The autocovariance function satisfies
(1) $\gamma(0) \geq 0$;
(2) $\gamma(-h) = \gamma(h)$;
(3) $|\gamma(h)| \leq \gamma(0)$.

Proof. [Supplement]

Step 1. $\gamma(0) = \mathrm{Var}(X_t) \geq 0$.

Step 2. $\gamma(-h) = \mathrm{Cov}(X_t, X_{t-h}) = \mathrm{Cov}(X_s, X_{s+h}) = \gamma(h)$ with $s = t - h$.

Step 3. Cauchy–Schwarz: $|\mathrm{Cov}(X_t, X_{t+h})| \leq \sqrt{\mathrm{Var}\, X_t \ \mathrm{Var}\, X_{t+h}} = \gamma(0)$. ■

Theorem 1.2.7 (Nonnegative definiteness)

For any reals $x_1, \ldots, x_k$ and integers $t_1, \ldots, t_k$,

$$\sum_{i,j=1}^{k} x_i \gamma(t_i - t_j) x_j \geq 0.$$

Proof. [Supplement] Let $Y = \sum_{i=1}^{k} x_i X_{t_i}$. Since $\mathrm{Cov}(X_{t_i}, X_{t_j}) = \gamma(t_j - t_i) = \gamma(t_i - t_j)$ by Theorem 1.2.6(2),

$$0 \leq \mathrm{Var}(Y) = \sum_{i,j=1}^{k} x_i x_j \,\mathrm{Cov}(X_{t_i}, X_{t_j}) = \sum_{i,j=1}^{k} x_i \gamma(t_i - t_j) x_j.$$

■

Thus $\Gamma_k = [\gamma(t_i - t_j)]_{i,j=1}^{k}$ is nonnegative definite; conversely, every even nonnegative definite function on $\mathbb{Z}$ is the ACVF of a stationary Gaussian series.

Definition 1.2.8 (White noise)

A stationary series $\{X_t\}$ whose ACVF satisfies $\gamma(h) = 0$ for $h \neq 0$ is called *white noise*, written $\{X_t\} \sim WN(0, \sigma^2)$ with $\sigma^2 = \gamma(0)$ (mean zero is understood). If in addition the X_t are i.i.d., we write $\{X_t\} \sim IID(0, \sigma^2)$.

Definition 1.2.9 (Autocorrelation function)

For a stationary $\{X_t\}$,

$$\rho(h) = \gamma(h)/\gamma(0)$$

is the *autocorrelation function* (ACF); for fixed h, $\rho(h)$ is the autocorrelation coefficient at lag h.

By Theorem 1.2.6, $\rho(0) = 1$, $\rho(-h) = \rho(h)$ and $|\rho(h)| \leq 1$.

Definition 1.2.10 (Multivariate normal distribution)

Let $\mu = (\mu_1, \ldots, \mu_k)^T \in \mathbb{R}^k$ and let $\Sigma = (\sigma_{ij})_{i,j=1}^{k}$ be symmetric positive definite. If the random vector $Z = (Z_1, \ldots, Z_k)^T$ has joint density

$$f(z) = \frac{1}{(2\pi)^{k/2} |\Sigma|^{1/2}} \exp\Big\{-\tfrac{1}{2}(z-\mu)^T \Sigma^{-1} (z-\mu)\Big\}, \qquad z \in \mathbb{R}^k,$$

then Z follows the *multivariate normal distribution* with mean μ and covariance Σ; here $\mu_i = EZ_i$ and $\sigma_{ij} = \mathrm{Cov}(Z_i, Z_j)$.

Definition 1.2.11 (Gaussian time series)

$\{X_t\}$ is a *Gaussian time series* if $(X_{t_1}, \ldots, X_{t_k})^T$ is multivariate normal for every k and all $t_1, \ldots, t_k$.

A normal law is determined by its mean and covariance, so a weakly stationary Gaussian series is strictly stationary. For a zero-mean stationary Gaussian series the best predictor is linear:

$$E(X_{k+1} \mid X_1, \ldots, X_k) = \sum_{i=1}^{k} a_i X_i,$$

with a_i estimable by minimizing $\sum_{t=k+1}^{n}(X_t - a_1X_{t-1} - \cdots - a_kX_{t-k})^2$.

[Supplement] Proof of linearity of the conditional mean. Let $\mathbf{X} = (X_1, \ldots, X_k)^T$, $\Gamma_k = \mathrm{Cov}(\mathbf{X})$ (invertible), $\gamma_k = \mathrm{Cov}(\mathbf{X}, X_{k+1})$, $a = \Gamma_k^{-1}\gamma_k$, $U = X_{k+1} - a^T\mathbf{X}$. Then $\mathrm{Cov}(\mathbf{X}, U) = \gamma_k - \Gamma_k a = 0$; as $(\mathbf{X}, U)$ is jointly normal, $U \perp \mathbf{X}$, so $E(U \mid \mathbf{X}) = 0$ and $E(X_{k+1} \mid \mathbf{X}) = a^T\mathbf{X}$. ■

Nonstationary data can often be reduced to stationarity by transformations such as differencing (Chapter 5).

1.3 Selected exercises

Motivation. Exercises on exponential smoothing, strict versus weak stationarity, Gaussian conditional means and sums of stationary series.

Exercise 1.1 (Exponential smoothing recursion)

Let $0 < w < 1$ and let S_t^* be defined by (1.3) from an initial value S_0^*. Show (1.4), i.e.

$$S_t^* = w\sum_{j=0}^{t-1}(1-w)^j x_{t-j} + (1-w)^t S_0^*, \qquad t \geq 1,$$

and deduce that $S_t^* - S_t \to 0$ as $t \to \infty$.

Solution. Induction on t.

Step 1. $t = 1$: (1.3) gives $S_1^* = wx_1 + (1-w)S_0^*$.

Step 2. $t \geq 2$: substituting the formula for $t-1$ into (1.3) and putting $k = j+1$,

$$\begin{aligned} S_t^* &= wx_t + (1-w)\Big\{w\sum_{j=0}^{t-2}(1-w)^j x_{t-1-j} + (1-w)^{t-1}S_0^*\Big\} \\ &= wx_t + w\sum_{k=1}^{t-1}(1-w)^k x_{t-k} + (1-w)^t S_0^* = w\sum_{k=0}^{t-1}(1-w)^k x_{t-k} + (1-w)^t S_0^*. \end{aligned}$$

Step 3. Hence $S_t^* - S_t = (1-w)^t S_0^* \to 0$ since $0 < 1-w < 1$. ■

Exercise 1.3 (Strict versus weak stationarity without independence)

Using examples that are *not* i.i.d., show that
(a) a strictly stationary series need not be weakly stationary;
(b) a weakly stationary series need not be strictly stationary.

Solution.

(a) Let Z be standard Cauchy and $X_t = Z$ for all t. Then

$$P(X_{t_1} \leq x_1, \ldots, X_{t_k} \leq x_k) = P(Z \leq \min_i x_i) = P(X_{t_1+h} \leq x_1, \ldots, X_{t_k+h} \leq x_k),$$

so $\{X_t\}$ is strictly stationary; it is not i.i.d. ($X_1 = X_2$, non-degenerate), and $EX_t^2 = \infty$ violates (2) of Definition 1.2.3.

(b) Let $\{X_t\}$ be independent with $X_t \sim N(0,1)$ for odd t and $P(X_t = \pm 1) = \frac{1}{2}$ for even t. Then $EX_t = 0$, $\mathrm{Cov}(X_t, X_{t+h}) = \mathbb{1}\{h = 0\}$, so $\{X_t\} \sim WN(0,1)$, but the marginals differ:

$$P(X_1 \leq \tfrac{1}{2}) = \Phi(\tfrac{1}{2}) \approx 0.691 \neq \tfrac{1}{2} = P(X_2 \leq \tfrac{1}{2}),$$

so it is neither i.i.d. nor strictly stationary. ■

Exercise 1.6 (Gaussian one-step conditional mean)

(a) (Exercise 1.5.) Let (X, Y) be bivariate normal with $\operatorname{Var}(Y) > 0$. Show that $E(X \mid Y) = aY + b$ with $a = \operatorname{Cov}(X, Y)/\operatorname{Var}(Y)$ and $b = EX - aEY$.
(b) Let $\{X_t\}$ be a stationary Gaussian series with mean μ and ACVF γ, $\gamma(0) > 0$. Express $E(X_t \mid X_{t-1})$ and its mean squared error through μ and γ.

Solution.

(a) Put $U = X - aY - b$.

Step 1. Every $sU + tY$ is a linear combination of X, Y, hence normal, so $Ee^{i(sU+tY)} = \exp\{i(sEU + tEY) - \frac{1}{2}\operatorname{Var}(sU + tY)\}$.

Step 2. $EU = 0$ and $\operatorname{Cov}(U, Y) = \operatorname{Cov}(X, Y) - a\operatorname{Var}(Y) = 0$, so $\operatorname{Var}(sU + tY) = s^2 \operatorname{Var} U + t^2 \operatorname{Var} Y$ and the characteristic function factorizes: $U \perp Y$.

Step 3. Hence $E(X \mid Y) = aY + b + E(U \mid Y) = aY + b$.

(b) (X_t, X_{t-1}) is bivariate normal with $\operatorname{Var} X_{t-1} = \gamma(0)$, $\operatorname{Cov}(X_t, X_{t-1}) = \gamma(1)$; (a) gives $a = \rho(1)$, $b = \mu - \rho(1)\mu$:

$$E(X_t \mid X_{t-1}) = \mu + \frac{\gamma(1)}{\gamma(0)}(X_{t-1} - \mu).$$

With $U = X_t - \mu - \rho(1)(X_{t-1} - \mu)$,

$$EU^2 = \gamma(0) - 2\rho(1)\gamma(1) + \rho(1)^2\gamma(0) = \gamma(0)(1 - \rho(1)^2) = \gamma(0) - \frac{\gamma(1)^2}{\gamma(0)}.$$

■

Exercise 1.7 (Sum of independent stationary series)

Let $\{X_t\}$ and $\{Y_t\}$ be independent stationary series with ACVFs γ_X, γ_Y. Show that $\{X_t + Y_t\}$ is stationary and find its ACVF.

Solution. Let $Z_t = X_t + Y_t$; check Definition 1.2.3.

Step 1. $EZ_t = \mu_X + \mu_Y$.

Step 2. $EZ_t^2 \le 2EX_t^2 + 2EY_t^2 < \infty$.

Step 3. By bilinearity,

$$\operatorname{Cov}(Z_t, Z_{t+h}) = \operatorname{Cov}(X_t, X_{t+h}) + \operatorname{Cov}(X_t, Y_{t+h}) + \operatorname{Cov}(Y_t, X_{t+h}) + \operatorname{Cov}(Y_t, Y_{t+h}).$$

Independence gives $\operatorname{Cov}(X_s, Y_u) = 0$, so

$$\gamma_Z(h) = \gamma_X(h) + \gamma_Y(h),$$

free of t (only uncorrelatedness was used), and

$$\rho_Z(h) = \frac{\gamma_X(0)\rho_X(h) + \gamma_Y(0)\rho_Y(h)}{\gamma_X(0) + \gamma_Y(0)}.$$

■